



These diamonds are hot

Fake diamonds helped scientists find hottest temperature ever recorded on earth
<http://digit.in/HotDiamonds>



Safari has security bugs

A new tool by the Project Zero team at Google revealed that Safari has many security bugs.
<http://digit.in/SafariBuggd>

Self-Diagnosis 2.0

Can tech help you get a few answers while you wait to see a doc?

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e're not going to mince words. Most doctors advise against self-diagnosis, because an

incorrect diagnosis can be fatal. Proper diagnosis is best left to a qualified specialist. The tools we list here are *never* to be considered a substitute for a real doctor. We only recommend them as a way to get some answers, or peace of mind whilst you wait for an appointment with a real medical practitioner. If you feel unwell, please seek proper medical attention immediately.

SMARTPHONE ONLY

SpiroSmart: This app uses the inbuilt microphone as a spirometer to monitor chronic lung ailments such as asthma,



cystic fibrosis and chronic bronchitis. You hold the phone at arm's length and exhale, and the sound of your breath is translated into a flow curve that is analysed for obstructions. Also see the iCare Lung capacity app.

PoopMD+: This free app reads stool colour for infants to identify biliary atresia (BA) – a disorder that causes paediatric liver disease. The color of a baby's poop can indicate a gastrointestinal illness or a problem with the liver, and early detection can greatly improve the outcome from this disease. BA is most common in East Asia,

with incidence of one in 5,000 live births. If left untreated BA can lead to liver failure in infants.

iCare Apps: iCare Fit Studio provides nine different apps on the Play Store, and their corresponding pro versions as separate apps, to form perhaps the most comprehensive mobile-only diagnostic ecosystem available. For instance, the iCare Eye Test app is a comprehensive solution for quite a number of eye tests that you can do on your own. The iCare hearing test app tests your ear's hearing and age with different frequencies, and it could check the degree of hearing loss in relation to the sound frequency.

DEVICE-BASED

Medical innovation isn't limited to just smartphones, but also use accessories to take things one step further.

Kardia Mobile: This is a device and app combo that measures blood pressure and plots an electrocardiogram (ECG) in order to detect hypertension and atrial fibrillation (AF). It's available for \$99, and is already FDA approved.

Biomeme System: The Biomeme device is a real-time PCR thermocycler that attaches to your iPhone SE and enables you to perform DNA analysis – effectively turning your iPhone into a mobile DNA laboratory. Using the lab-free sample prep (there are tons of options available on their store depending on what you're trying to detect), you isolate DNA from your environment, add it to a specific test cartridge and put it into the thermocycler. In less than an hour, you can determine the presence or absence of that target in your sample. It's like a targeted copy machine for DNA or RNA. If there's just one copy of your DNA target in your sample, the device turns it into billions of copies, which can then be detected. The iPhone is used to run the hardware and process the results. 

Retinoblastoma in kids

Retinoblastoma is a rare but aggressive form of eye cancer found almost exclusively in children. It can be very damaging to the child's vision, and even fatal, unless detected early. You could identify it early with a simple trick. Just click a picture of the child from your smartphone with the flash on. If any eye shows a white spot, make an appointment with a doctor immediately to rule out retinoblastoma.

Cardio Rhythm

MIT Media Lab spinout startup Cardiio has developed a mobile app that can use the camera on a smartphone to detect facial signs of heart arrhythmia, that could be indicative of a stroke. Enhancing on their already established app Cardiio, that can detect heartbeat from facial light reflection, Cardiio Rhythm uses the same data to detect irregular heartbeat patterns of atrial fibrillation (AF). Currently the app is being used for research only and has been validated by several academic papers. In the near future, Cardiio hopes to make the app available to consumers for early heart-arrhythmia screening.